

InSpect and AutoSpect GUI User Manual

Installation, automated processing, spectral review, calibration, and image export

Portrait-layout edition with standardized figure callouts and control tables

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This manual describes the InSpect interface and the AutoSpect processing workflow for LA-ICP-TOF-MS imaging data. It also documents RetroSpect, the Isotope Selection GUI, ProSpect, concentration-calibration tools, and the tiled Images view.

Quick Start

The fastest complete workflow is:

1. Install MATLAB Runtime R2017a and MATLAB Runtime Update 3.
2. Install the latest compiled InSpect release and launch InSpect.
3. Select Programs → Launch AutoSpect.
4. Use the default ion-selection file unless a validated custom selection is available.
5. Select individual raw files or a directory, then confirm the files to process.
6. Review and accept the empirical tailing parameters.
7. Review and accept the instrumental-notching parameters.
8. Allow AutoSpect to complete peak-profile generation and spectral fitting.
9. Review drift correction, fitted spectra, calibration models, and elemental maps in InSpect.
10. Save the InSpect session and export images or plots as needed.

Input Data Requirements

- Raw data are HDF5 (.h5) files generated by TOFWERK TofPilot for supported S2, R, and 2R instruments.
- AutoSpect writes fitted information to MATLAB (.mat) output and updates the PeakData structure in the source HDF5 file.
- Custom ion-selection files and RetroSpect parameter files should be used only when they have been validated for the current instrument tuning.
- Concentration calibration requires fitted reference-standard data and the corresponding reference values.

Important: AutoSpect tailing and notching parameters are instrument-tuning dependent. Reuse them only for datasets acquired under comparable tuning conditions.

1. Software Overview and Installation

AutoSpect automatically processes and fits laser ablation inductively coupled plasma time-of-flight mass spectrometry (LA-ICP-TOF-MS) imaging data. It performs drift correction, peak-profile modeling, spectral fitting, and quantitative elemental image reconstruction. InSpect provides the main environment for reviewing and refining these results.

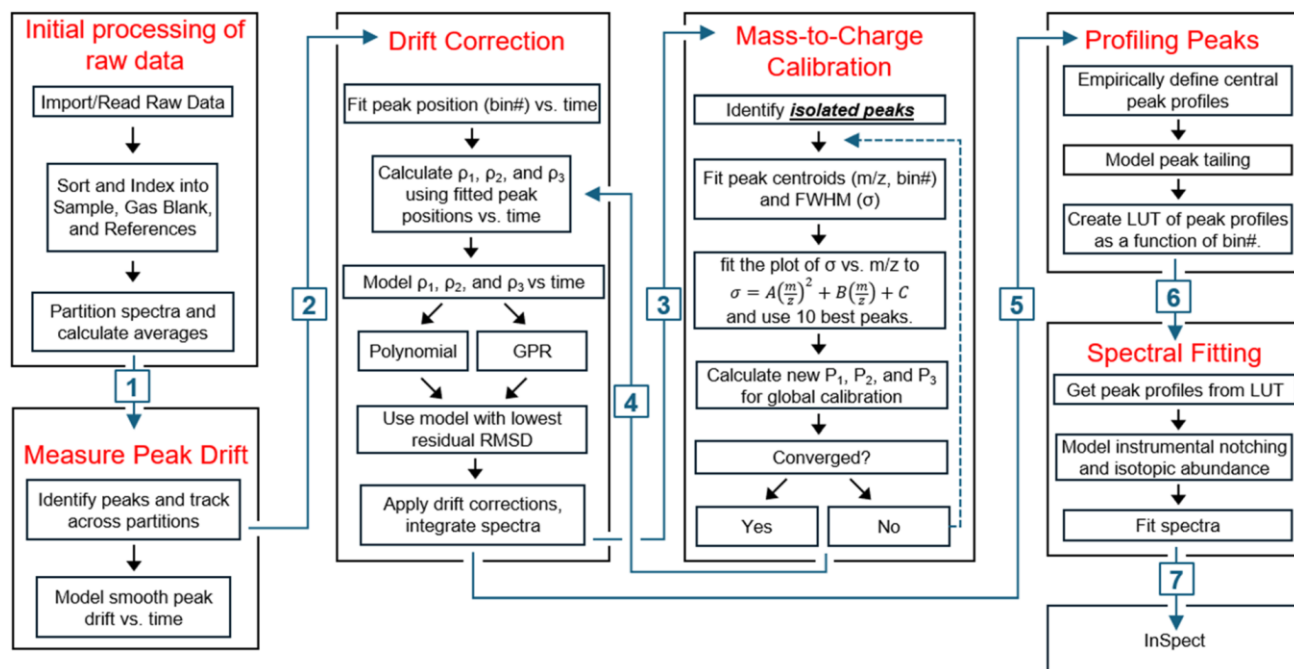


Figure 1. High-level AutoSpect processing workflow, from raw-data import through drift correction, calibration, peak profiling, spectral fitting, and downstream review.

1.1 Software Components

Item	Description
InSpect	Main graphical interface for loading fitted data, reviewing drift correction and spectral fits, building concentration calibrations, inspecting images, and launching supporting tools.
AutoSpect	Core semi-automated processing and fitting pipeline for raw LA-ICP-TOF-MS imaging data.
RetroSpect	Review interface for empirical peak-tail and instrumental-notch parameters used during AutoSpect fitting.
ProSpect	Directory-monitoring utility that detects new .h5 files and launches AutoSpect automatically.
ReSpect	A supporting component of the software suite; detailed operation is outside the scope of this edition.

1.2 Install MATLAB Runtime

The compiled Windows version of InSpect requires MATLAB Runtime R2017a and MATLAB Runtime Update 3.

1. Download and install MATLAB Runtime R2017a: [R2017a Windows installer](#)
2. Download and install MATLAB Runtime Update 3: [R2017a Update 3](#)

1.3 Install InSpect

Download the current compiled Windows release from the InSpect repository: [AutoSpect compiled InSpect releases](#)

- Extract the downloaded archive.
- Place the extracted program in the preferred software directory.
- Optionally create a desktop shortcut to the executable.

1.4 Platform Availability

Item	Description
Windows	Available. This manual describes the Windows workflow.
Linux	Not currently available in the compiled distribution.
macOS	Not currently available in the compiled distribution.

1.5 Data Storage and Access

Item	Description
AutoSpect processing	Reads raw HDF5 files produced by ToFPilot. Processed variables are stored in MATLAB format, and fitted information is used to update the PeakData structure in the source HDF5 file.
InSpect sessions	Stores GUI state and processed data in MATLAB (.mat) format and may also update data stored in the TOFWERK HDF5 file.
Elemental mapping: .mat	InSpect can load fitted MATLAB files saved by AutoSpect.
Elemental mapping: .h5	InSpect reads supported TOFWERK HDF5 data from S2, R, and 2R instruments.

2. Process Your First Dataset

The following workflow describes a typical new-data processing session. The AutoSpect prompts appear in sequence and return the user to InSpect after fitting is complete.

2.1 Launch AutoSpect

Launch InSpect, then select Programs → Launch AutoSpect from the top menu.

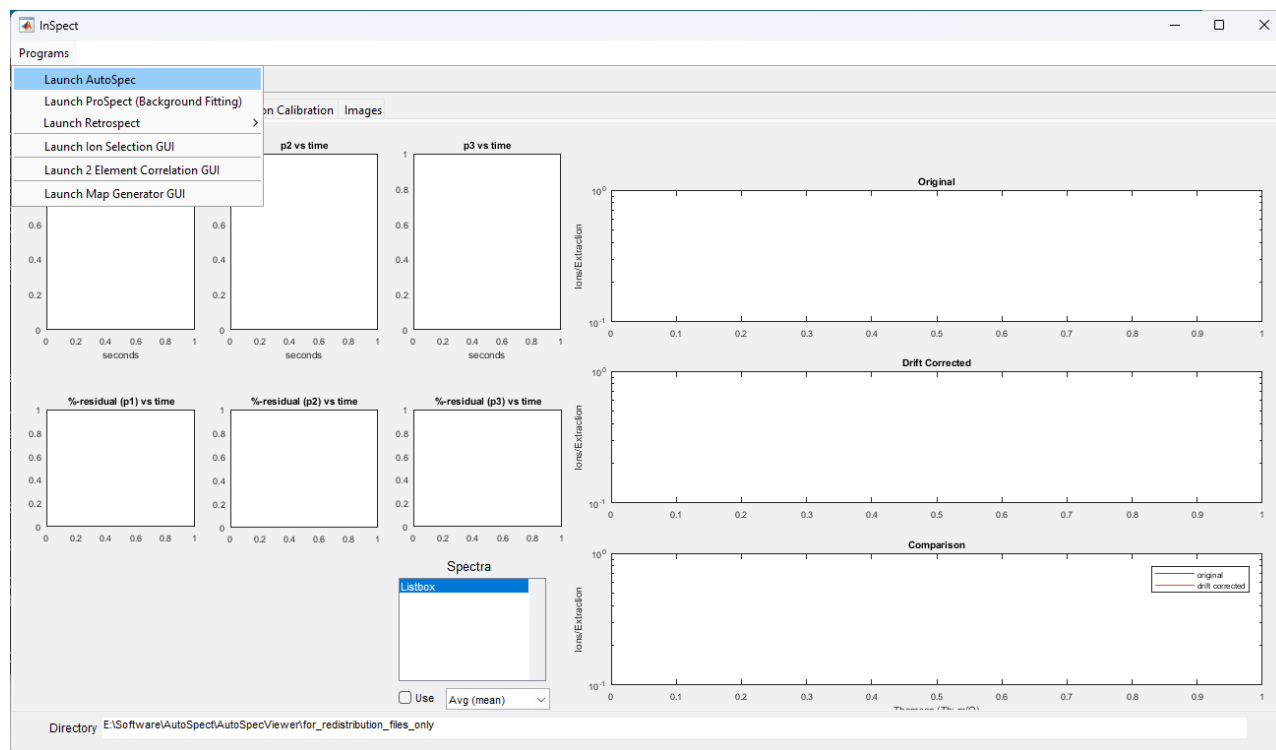


Figure 2. Launching AutoSpect from the InSpect Programs menu.

2.2 Select an Ion-Selection File

AutoSpect asks whether a nondefault ion-selection file should be loaded. Select No unless a previously validated custom ion-selection file is required for the current dataset.

2.3 Select Files for Processing

Choose whether to select individual files or an entire directory. The file browser reopens after each selection so that additional files or folders can be added. Select Cancel in the file browser after all desired data have been added.

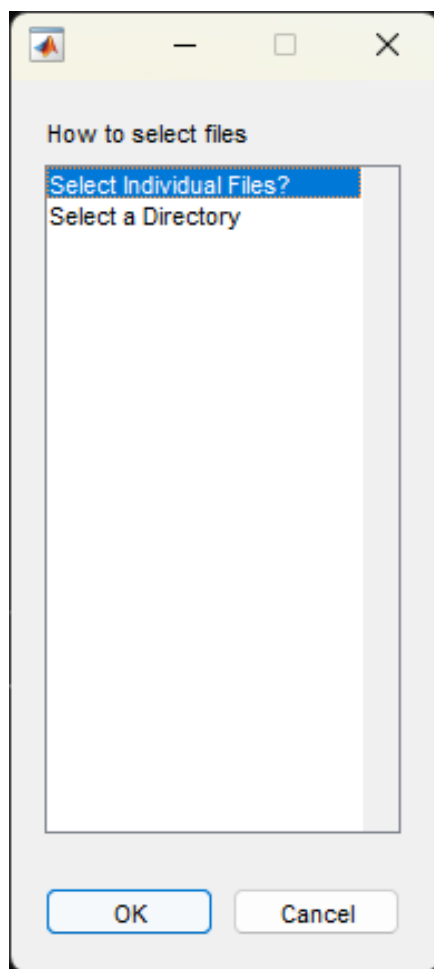


Figure 3. File-selection mode dialog for individual files or an entire directory.

2.4 Confirm the Selected Files

AutoSpect displays the selected files for confirmation before preprocessing begins. This provides a final opportunity to verify that the correct subset of a larger directory was selected.

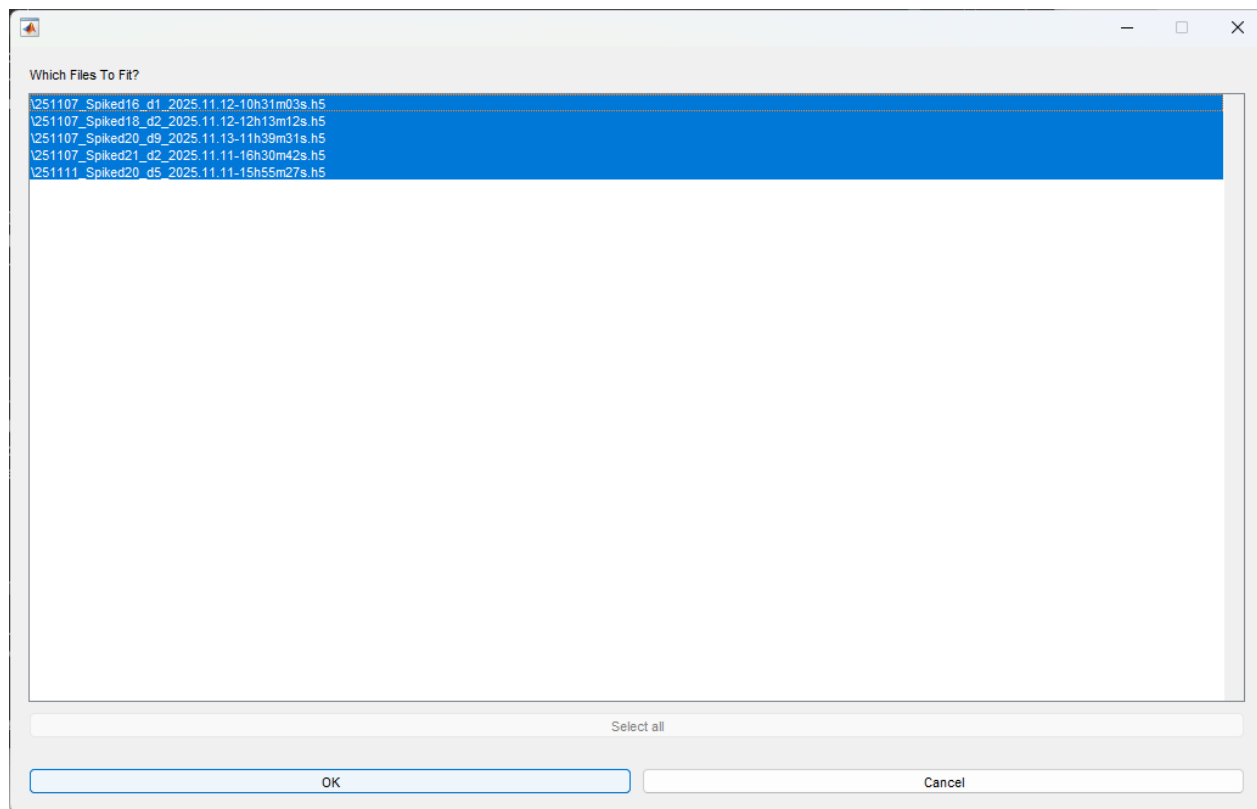


Figure 4. Selected-file confirmation dialog.

2.5 Automatic Preprocessing

AutoSpect performs initial preprocessing, including drift correction and partitioning of the raw spectral data. When preprocessing is complete, the program opens the RetroSpect review interface for the empirical tailing and instrumental-notching parameters.

2.6 Review Tailing Parameters

AutoSpect models instrument-specific peak tails before fitting. The fitted spectrum should closely overlap the background-corrected spectrum across the mass range, while the modeled high- and low- m/z tails follow the local baseline without extending through peak centroids.

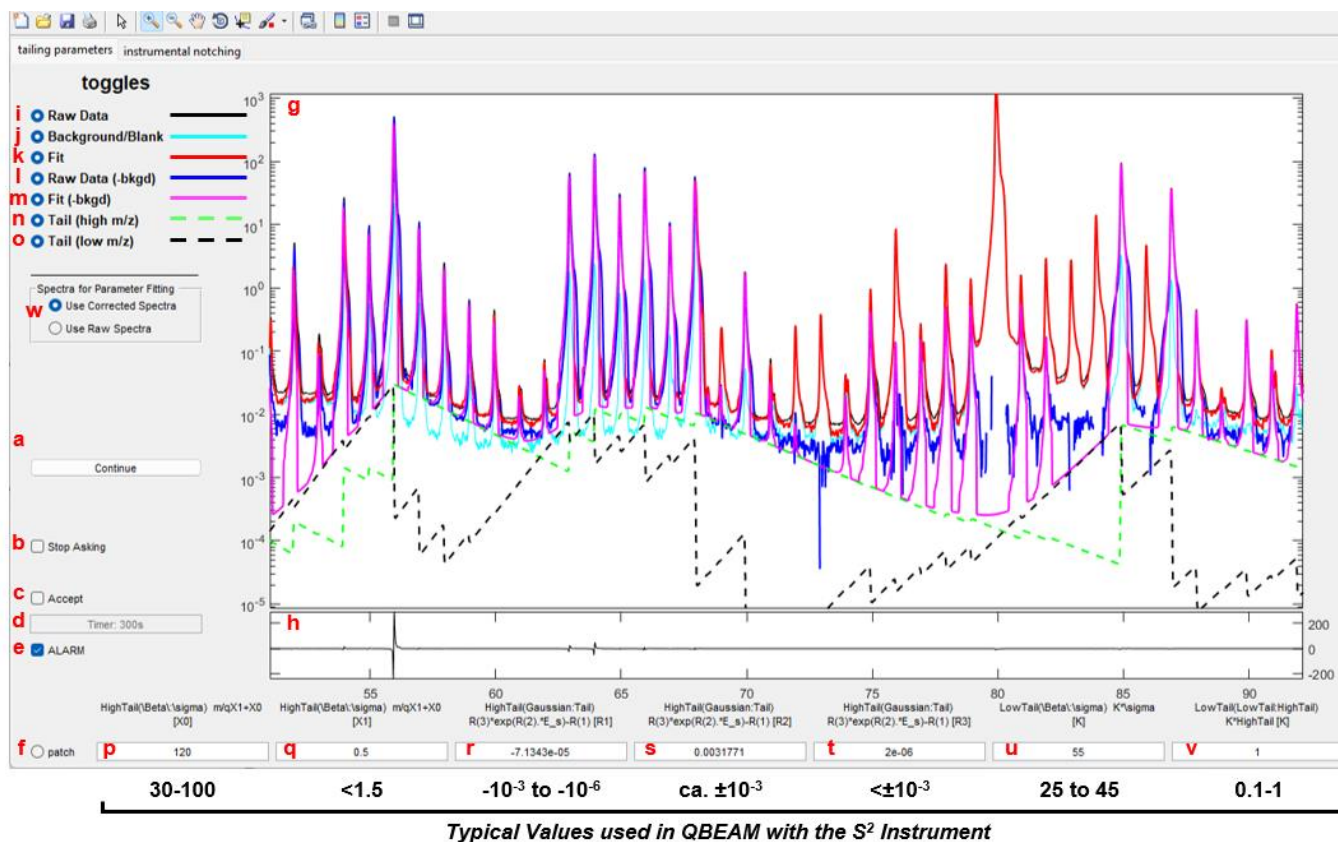


Figure 5. RetroSpect Tailing Parameters interface. The main plot shows raw data, background/blank, fitted spectra, corrected spectra, and the high- and low-m/z tail components.

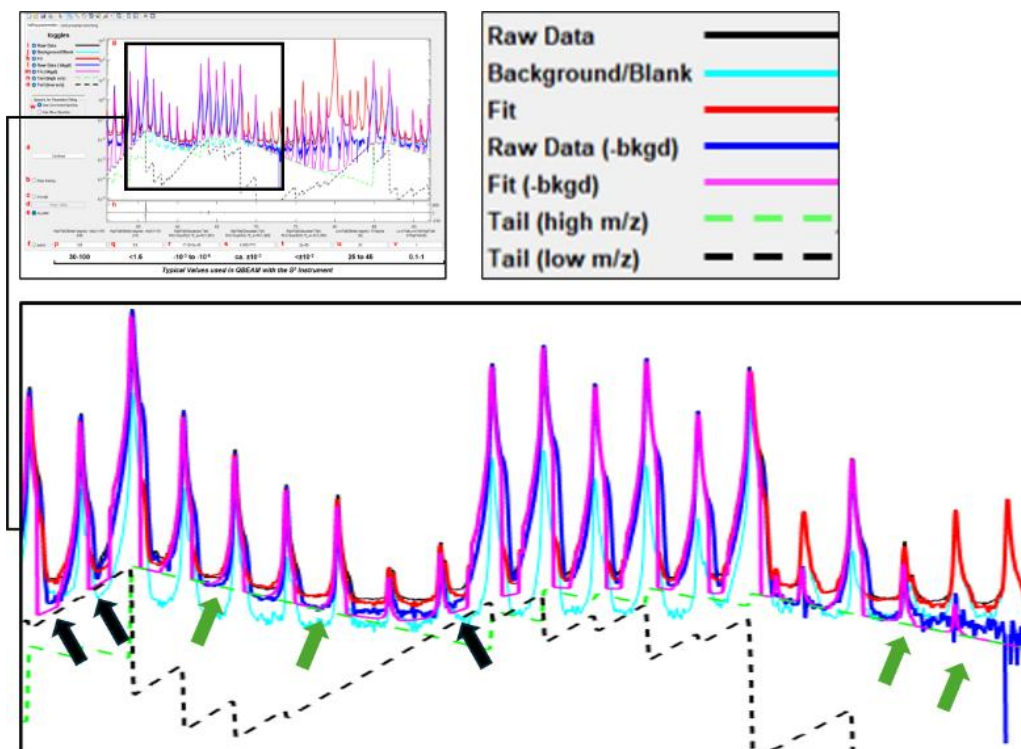


Figure 6. Expanded tailing example. Black dashed lines indicate the low-m/z tail and green dashed lines indicate the high-m/z tail.

1. For the low-m/z leading tail, adjust parameters r, s, and t so that the dashed black tail follows the corrected baseline without crossing the peak itself.
2. For the high-m/z trailing tail, adjust parameters u and v so that the dashed green tail follows the corrected baseline.
3. Select Accept to retain the displayed parameters.
4. Select Stop Asking when the same validated parameters should be reused for the remaining datasets in the current batch.

5. Select Continue to proceed to the instrumental-notching review.

Note: Click Continue only once. Multiple clicks can be queued and may automatically accept the default notching parameters on the following screen.

2.7 Review Instrumental Notching

Instrumental notches are narrow regions of signal suppression. Adjust notch position and amplitude so that modeled isotope ratios agree with the empirical centroid ratios in the raw spectrum. For example, related argon isotopes should retain the expected relative response after correction.

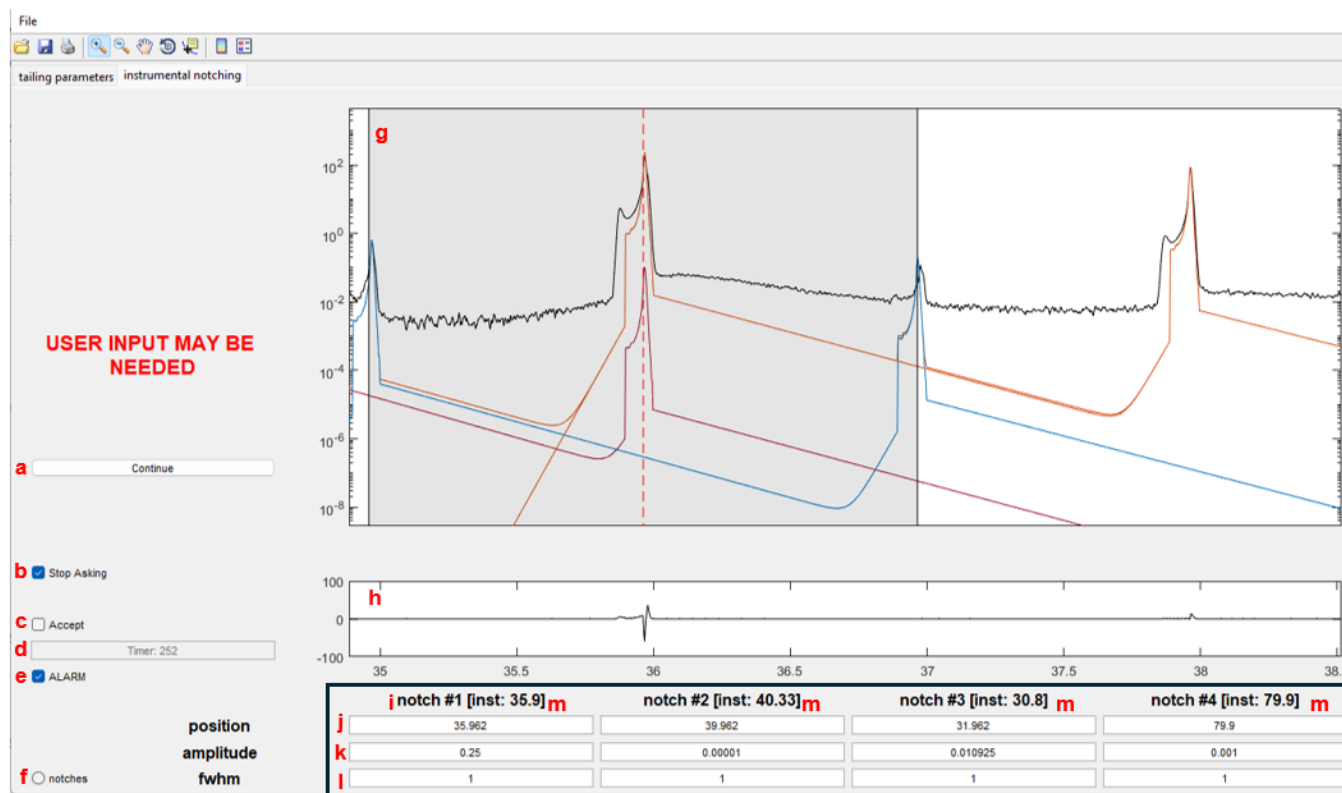


Figure 7. RetroSpect Instrumental Notching interface. Shaded regions and dashed centers show the notches being parameterized.

1. Adjust the notch position to align the modeled suppression with the affected m/z region.
2. Adjust amplitude between 0 and 1, where 1 applies no notching and smaller values apply progressively stronger suppression.
3. Retain the current FWHM value unless the fitting workflow specifically requires a change.
4. Select Accept, optionally select Stop Asking for batch reuse, and then select Continue.

2.8 Automatic Spectral Fitting and Return to InSpect

After tailing and notching are accepted, AutoSpect generates the peak profile and fits the selected spectra automatically. When processing is complete, InSpect reappears. If no processed file was already loaded, the final fitted file is loaded automatically.

3. InSpect Main Interface

InSpect is the primary interface for inspecting fitted data. The main tabs documented here are Drift Correction, Spectral Fitting, Concentration Calibration, and Images.

3.1 Drift Correction Tab

The Drift Correction tab summarizes the fitted drift parameters, residuals, and the effect of correction on the spectra. Plot interaction tools include zoom, pan, and data-cursor modes; these modes temporarily disable plot context menus.



Figure 8. Drift Correction tab. Callouts identify file/session controls, drift-parameter plots, spectral comparisons, spectrum-selection controls, program launchers, and context menus.

Callout	Control / Region	Function
1a	Directory	Displays the directory for the loaded scan.
1b	Open	Loads AutoSpect-fitted data or an InSpect session.
1c	Save	Saves the current InSpect session and amends the fitted file.
1d-1f	Zoom / data cursor / pan	Toggles plot navigation modes. Context menus are disabled while these tools are active.
2a-2c	P1-P3 drift plots	Shows the measured and fitted drift-correction parameters as a function of time.
2d-2f	P1-P3 residuals	Shows the residuals of the corresponding drift fits.
3a	Original	Plots uncorrected spectra for the data partitions.
3b	Drift Corrected	Plots drift-corrected spectra for the same partitions.
3c	Comparison	Compares average original and drift-corrected spectra.

The screenshot displays the InSpec software interface, which is used for spectral analysis. The main window is titled "InSpec (c) Dr. Andrew M. Crawford: [LA_Brain_2A_20um_xy20_LP70_100Hz_2025-05-12_15h32m47s_Fitted_w_AutoSpec.mat]".

The interface is divided into several sections:

- Top Panel:** Contains buttons for "Drift Correction", "Spectral Fitting", "Concentration Calibration", and "Images".
- Left Panel (Color Map):**
 - Color Map:** Set to "bik2jet".
 - Combine:** Set to "b".
 - Aspect Ratio:** Set to "aspect ratio (constr...)".
- Main Display Area:**
 - Top Plot (Integrated Spectrum Fit):** Shows a 3D plot of intensity versus position (x, y, z). The plot is labeled "2" and "c". It displays multiple overlapping spectra with peaks labeled "c" and "2c".
 - Bottom Plot (Image ROI Spectrum):** Shows a 2D plot of intensity versus position (x, y). The plot is labeled "d" and "2d". It displays a single spectrum with peaks labeled "d" and "2d".
- Right Panel (Spectral Fitting Parameters):**
 - Buttons:** "Raw", "Baseline/Blank", "Fit", "Raw [-bkgd]", "Fit [-bkgd]", "Selected", "Centers", "All Spectral Components".
 - Parameters:**
 - Sample:** "Sample" (dropdown).
 - Assigned Image ROIs:** "Integrated" (dropdown).
 - Isotope /:** "Zn+1" (dropdown).
 - Normalization:** "Li+1" (dropdown).
 - High:** "0.99" (dropdown).
 - Low:** "0.01" (dropdown).
 - Auto Adjust X:** "Auto Adjust X" (radio button).
 - use:** "use" (radio button).

Diagram showing three boxes with red asterisks above them:

- Box 1: **a** ROI(add), **b** export plot
- Box 2: **c** list ions/isotopes, **d** export plot
- Box 3: **e** change font size

Callout	Control / Region	Function
0a	Directory	Displays the parent directory.
0b / 0c	Open / Save	Loads or saves fitted MATLAB data.

Callout	Control / Region	Function
0d-0g	Zoom / data cursor / pan / rotate	Controls plot navigation and 3D rotation.
1a	Colormap	Chooses the colormap applied to the displayed image.
1b	Combine	Applies N×N neighboring-pixel averaging for visualization only.
1c	Aspect ratio	Constrains the displayed map to its physical aspect ratio.
1d	Image map	Displays the mapped isotope, element, or species.
1e	Spectra	Selects Sample, Gas Blank, or a reference measurement.
1f	Assigned Image ROI	Selects a user-assigned region of interest.
1g	Isotope / Element	Selects the fitted species shown in the image and spectrum.
1h / 1j	Normalization / Use	Selects and enables an isotope or element used for normalization.
1i	Auto Adjust X	Automatically centers the spectral x-axis around the selected species.
1k / 1l	High / Low	Selects upper and lower quantile-based image display limits.
1m / 1n	Custom high / low	Accepts absolute display limits and switches the corresponding selector to Custom.

3.2.2 Spectral Display and Context Menus

Callout	Control / Region	Function
2a	Integrated Spectrum Fit	Legacy tab retained for compatibility; Image ROI Spectrum is preferred for routine review.
2b	Image ROI Spectrum	Displays the fit for the selected Assigned Image ROI.
2c	Spectrum and fit	Shows the measured spectrum and selected fit components.
2d	Vectorized intensities	Shows the pixel intensities from the mapped image as a vector.
2e	Raw	Toggles the raw spectrum.
2f	Baseline / Blank	Toggles the baseline or blank spectrum.
2g	Fit	Toggles the model fit to the raw spectrum.
2h	Raw [-bkgd]	Toggles the background-corrected raw spectrum.
2i	Fit [-bkgd]	Toggles the background-corrected fit.
2j	Selected	Toggles the fitted profile of the selected species.
2k	Centers	Toggles fitted centroid positions.
2l	All Spectral Components	Toggles all fitted species components.
3a	ROI (add)	Draws a polygon on the map or selects points from vectorized data to define an ROI.
3b / 3d	Export plot	Exports the corresponding plot.
3c	List ions/isotopes	Selects an m/z interval and lists possible monoatomic and polyatomic species.
3e	Change font size	Changes the fixed GUI font size.

3.3 Concentration Calibration Tab

The Concentration Calibration tab supports raster filtering, forward calibration surface fitting, inverse mapping from signal to concentration, conventional 2D calibration review, and application of the selected model to the loaded dataset.

3.3.1 Raster Filtering

Raster Filtering displays the signal behavior within each calibration raster so that unstable regions, transient responses, and anomalous pixels can be identified before surface fitting. Review the per-element raster panels and the selected-raster trace before proceeding to the calibration surface.

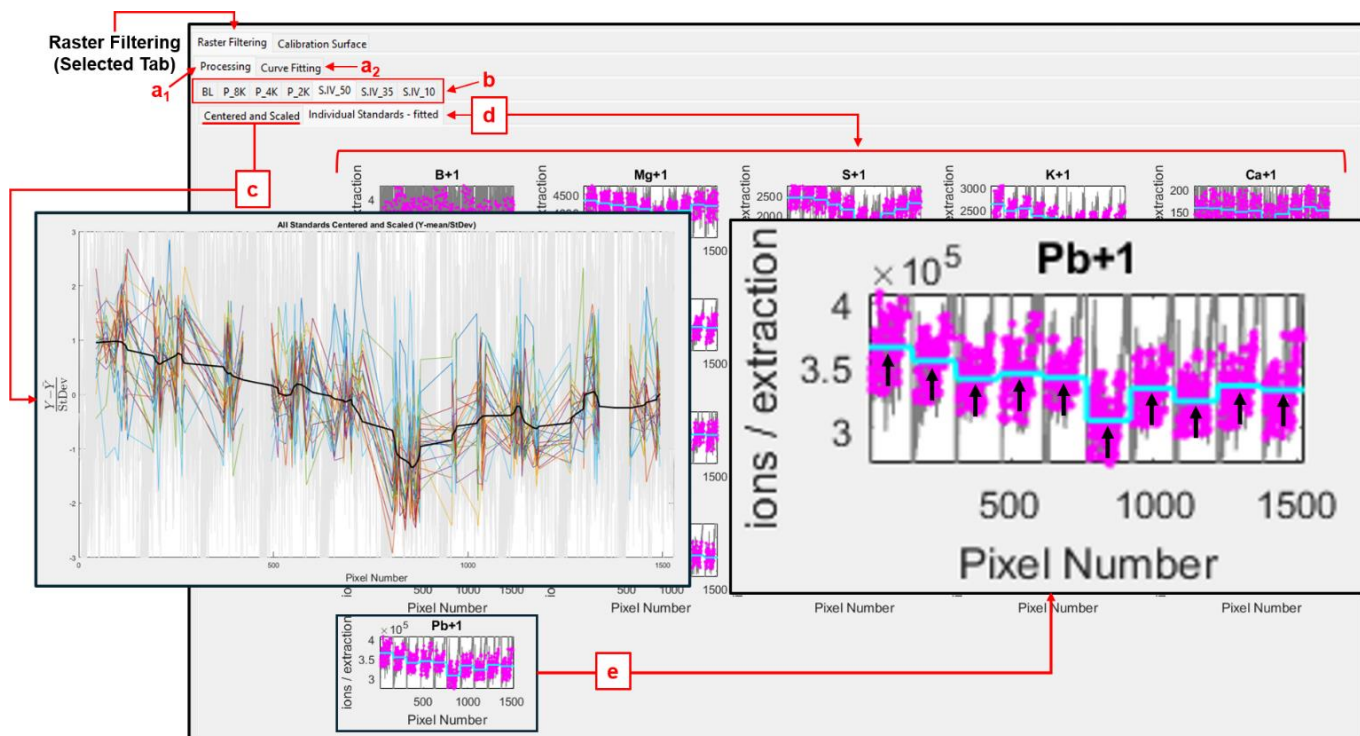


Figure 10. Raster Filtering view. The interface combines the selected-raster trace with per-species raster panels and highlighted extraction regions.

Callout	Control / Region	Function/Description
a ₁	Processing tab	Displays the filtered raster data
a ₂	Curve Fitting Tab	Currently, this does nothing. Reserved for future implementation.
b	Tabs for the Individual measured calibrants	These switch between the different calibrants that were collected inline with the samples. Each tab has its own associated tabs for (c) "Centered and Scaled" and (d) "Individual Standards – fitted"
c	Centered and Scaled tab	Each calibrant series (e.g., Fe+ all concentrations) is centered and scaled and plotted overlayed with all other calibrants.
d	Individual Standards – Fitted tab	Each calibrant raster (grey data) is filtered to reject outliers, the filtered data are retained (magenta data points), and the mean value (cyan line through each raster) is retained and carried forward for 3D mass calibration surface calculations in later sections.
e	Enlarged individual panel	An enlarged panel showing detail of the calibrant rasters for Pb ⁺¹ (grey data) which underwent outlier rejection (magenta circles), and the mean value (cyan line through each raster emphasized by the black arrows) which was retained and carried forward for 3D mass calibration surface calculations in later sections.

3.3.2 Forward Calibration Surface

The Forward View displays the modeled signal surface $S(T,M)$ as a function of acquisition time and concentration or mass. It provides controls for fitting method, smoothing, zero anchoring, residual display, GPU/FastGrid acceleration, and raster/series/calibrant exclusion.

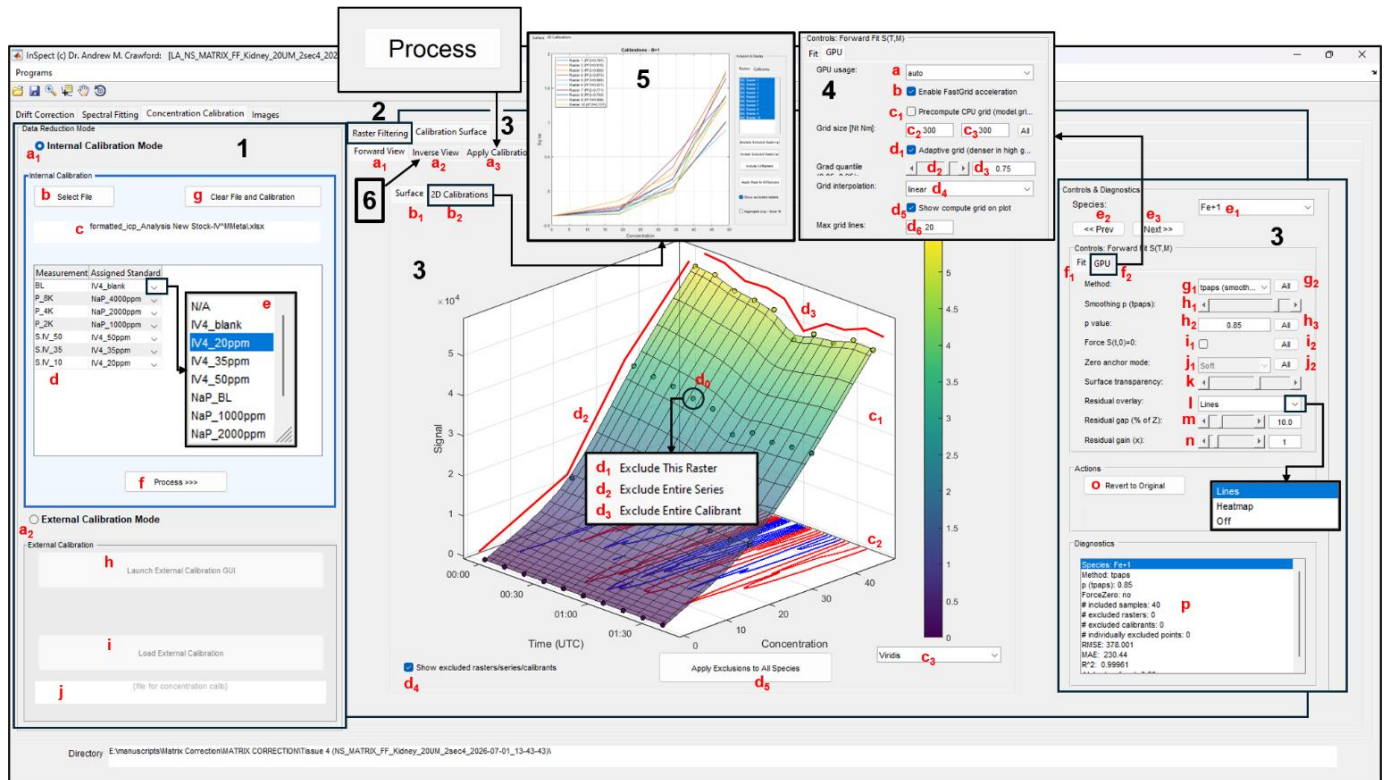


Figure 11. Forward calibration surface. Callouts identify the data-reduction mode, calibration assignments, surface display, fitting controls, GPU controls, diagnostics, and exclusion workflow.

Callout	Control / Region	Function
1a / 1a2	Internal / External Calibration Mode	Selects the calibration data-reduction approach.
1b-1e	Calibration file and assignments	Loads the reference-value file, lists measurements, and assigns each measurement to a standard.
1f	Process	Generates the forward and inverse calibration models.
1g	Clear File and Calibration	Clears the current reference data and generated surfaces.
1h-1j	External calibration controls	Launches the external calibration builder or loads a previously consolidated external calibration.
3a1-3a3	Forward / Inverse / Apply Calibration	Switches between the forward surface, inverse surface, and final apply-calibration workflow.
3b1 / 3b2	Surface / 2D Calibrations	Switches between the 3D surface and conventional 2D calibration curves.
3c1-3c3	Surface, residuals, colormap	Displays the model, residual projection, and colormap selector.
3d0-3d3	Right-click exclusions	Excludes the selected raster, entire series, or entire calibrant.
3d4 / 3d5	Show excluded / Apply to all species	Displays excluded data and optionally copies exclusions to every species.
3e1-3e3	Species / Prev / Next	Selects the current species or steps through the species list.
3f1 / 3f2	Fit / GPU	Switches between fitting and GPU/FastGrid controls.
3g1 / 3g2	Method / apply to all	Chooses tpaps or interpolant fitting and can apply the choice to all species.
3h1-3h3	Smoothing	Adjusts forward-surface smoothing and optionally applies the value to all species.
3i1 / 3i2	Force $S(t,0)=0$	Adds a zero-concentration anchor and can apply the choice to all species.
3j1 / 3j2	Zero Anchor Mode	Selects Soft or Hard anchoring and can apply the choice to all species.
3k-3n	Transparency and residual display	Controls surface transparency, residual mode, residual gain, and residual gain.
3o	Revert to Original	Restores the current species to its initial settings.
3p	Diagnostics	Reports species, method, smoothing, fit quality, and included/excluded counts.

Choosing a Surface-Fitting Method

Thin-Plate Smoothing Spline (tpaps). Constructs a smooth continuous surface that balances agreement with the measured calibration points against surface curvature. Higher smoothing-parameter values follow the data more closely, whereas lower values suppress local noise and produce a smoother model. Because the inverse calibration is generated numerically from the forward surface, tpaps is recommended for most datasets.

Interpolant. Constructs an exact scattered-data interpolation that passes through the measured calibration values without smoothing. It preserves genuine local features but also reproduces experimental noise and outliers. A single erroneous point can generate local peaks, valleys, or nonmonotonic behavior that destabilizes the inverse model.

Hint: Use tpaps for most calibration datasets. Reserve Interpolant for highly reproducible calibration measurements where exact reproduction of each point is required.

Zero Anchoring and Diagnostics

Force $S(t,0)=0$. Constrains the calibration response toward zero signal at zero analyte concentration. Enable this only when a zero-concentration response is physically justified after background correction.

Soft anchor. Treats the origin as a heavily weighted point and allows a small deviation when needed to preserve a smooth, stable surface. This is generally the preferred option.

Hard anchor. Forces the surface to pass exactly through the origin. This strict constraint can introduce curvature or distortion when the measured data are inconsistent with a zero intercept.

Residual overlay. Displays the difference between the measured calibration data and the fitted surface as lines, a heatmap, or an off state. Structured residual patterns can indicate a poor model, inappropriate smoothing, or anomalous calibration measurements.

GPU and Computational Grid Controls

- GPU usage selects automatic, CPU, or GPU evaluation when compatible hardware is available.
- FastGrid acceleration precomputes the concentration-time grid used for surface evaluation.
- Grid Size [Nt Nm] controls the time and concentration resolution of the computational grid.
- Adaptive grid refinement allocates more grid points to regions with high curvature or rapid signal change.
- Linear grid interpolation is recommended for most calibration surfaces. Nearest preserves discrete regions but may produce visible discontinuities.
- Show Compute Grid on Plot and Maximum Grid Lines affect visualization only; they do not alter the fitted calibration model.

3.3.3 Inverse View

The Inverse View displays $m(t,s)$, the model used to convert measured signal and acquisition time back to estimated concentration. Review smoothness, monotonicity, extrapolation, residuals, and the usable signal range before applying the calibration.

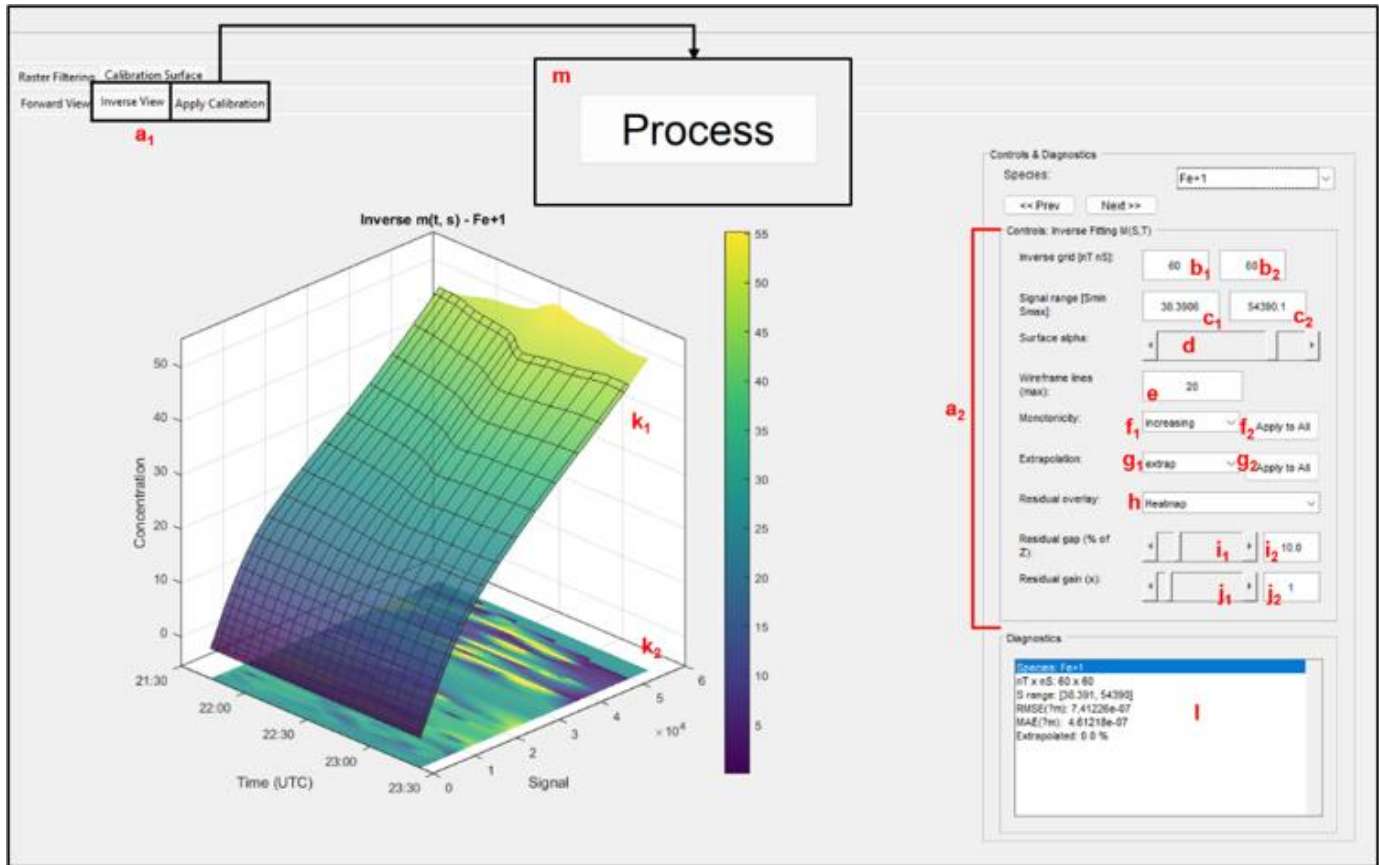


Figure 12. Inverse calibration surface and associated controls.

Callout	Control / Region	Function
a1	Inverse View tab	Displays the inverse model $m(t,s)$.
a2	Inverse Fitting panel	Contains inverse-grid, signal-range, monotonicity, extrapolation, and residual controls.
b1 / b2	Inverse grid [nT nS]	Sets the time- and signal-grid resolution used to evaluate and draw the inverse model.
c1 / c2	Signal range [Smin Smax]	Sets the signal range used for inverse fitting and display.
d	Surface alpha	Adjusts inverse-surface transparency.
e	Wireframe lines	Limits the number of wireframe lines displayed.
f1 / f2	Monotonicity / Apply to All	Selects monotonic behavior and optionally applies it to all species.
g1 / g2	Extrapolation / Apply to All	Selects out-of-range behavior and optionally applies it to all species.
h	Residual overlay	Chooses lines, heatmap, or off.
i1 / i2	Residual gap	Offsets the residual projection below the inverse surface.
j1 / j2	Residual gain	Scales the residual projection.
k1	Inverse surface	Shows estimated concentration as a function of time and signal.
k2	Residual projection	Shows inverse-model mismatch below the surface.
l	Diagnostics	Reports grid size, signal range, inverse error, and extrapolated fraction.
m	Process	Sends the reviewed calibration model to the downstream processing pipeline.

1. Select the species to review.
2. Inspect the inverse surface for smooth and physically reasonable behavior.
3. Adjust the signal range when the display does not cover the relevant calibrated interval.
4. Set monotonicity and extrapolation behavior, then apply settings to all species only when appropriate.
5. Review the residual projection and diagnostics before selecting Process.

3.3.4 2D Calibrations View

The 2D Calibrations view shows calibration curves by series or calibrant and provides list-based inclusion and exclusion controls. It is useful for identifying inconsistent series or standards before the final surface is accepted.

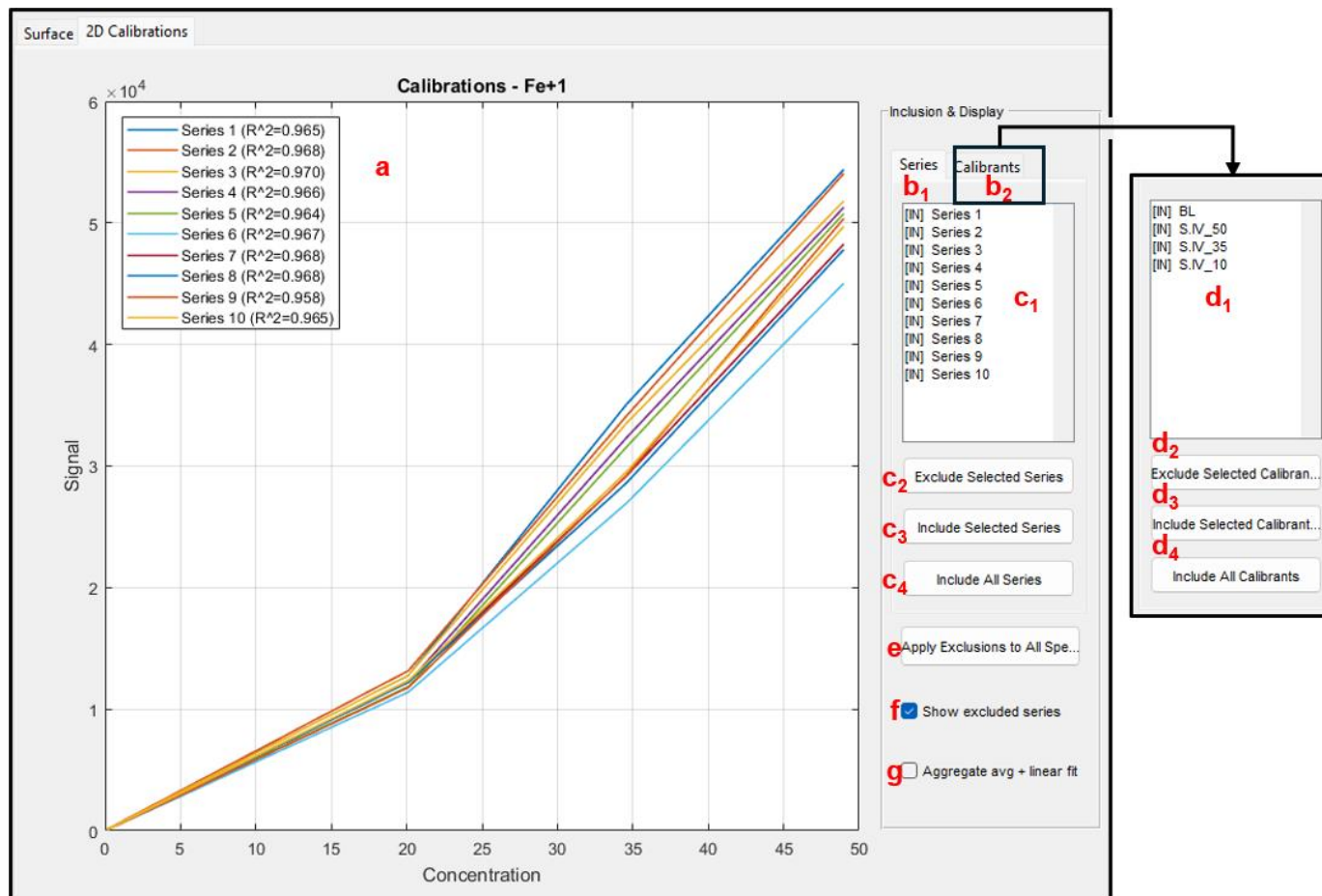


Figure 13. 2D calibration curves with series- and calibrant-level inclusion controls.

Callout	Control / Region	Function
a	2D calibration plot	Displays the calibration curves for the included series or calibrants.
b1 / b2	Series / Calibrants tabs	Switches the list between series-level and calibrant-level controls.
c1	Series list	Lists the series and their current inclusion status.
c2-c4	Series inclusion controls	Excludes selected series, includes selected series, or restores all series.
d1	Calibrant list	Lists calibrants and their current inclusion status.
d2-d4	Calibrant inclusion controls	Excludes a selected calibrant, restores it, or restores all calibrants.
e	Apply Exclusions to All Species	Copies the current series and calibrant exclusions to every species.
f	Show excluded series	Toggles display of excluded series.
g	Aggregate avg + linear fit	Displays the aggregate average and linear-fit information when enabled.

3.3.5 Apply Calibration

After the forward and inverse models have been reviewed, open Apply Calibration and select Process to apply the selected calibration models to the loaded image dataset.

3.4 Images Tab

The Images tab displays selected 2D isotope, element, or species maps in a tiled, scrollable layout. It supports display scaling, masking, orientation changes, linked navigation, despeckling, scale bars, and journal-style TIFF export.

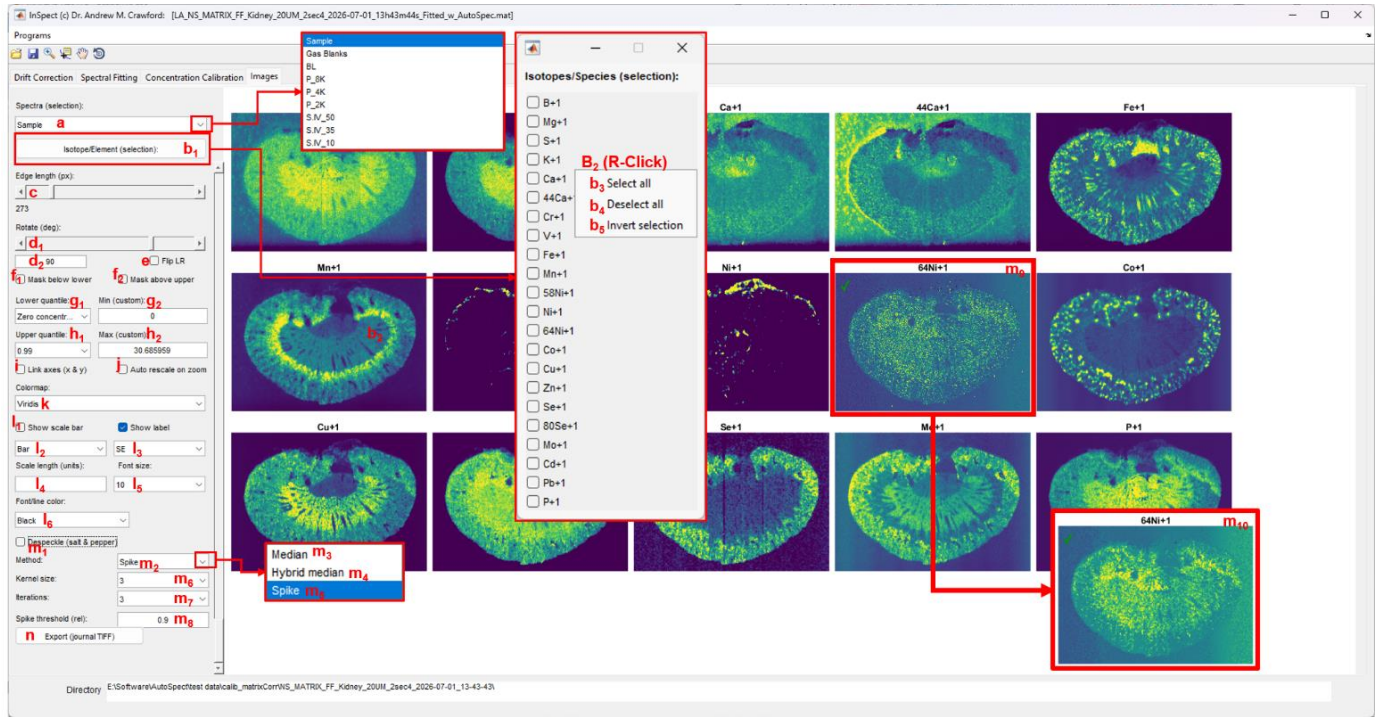


Figure 14. Images tab. The species selector, tiled image viewer, active-image indicator, display controls, despeckling controls, scale-bar controls, and export function are identified.

Callout	Control / Region	Function
a	Dataset selector	Chooses the dataset displayed in the image viewer.
b1 / b2	Isotope/Element selector	Opens the species-selection popover and lists the available maps.
b3-b5	Select all / Deselect all / Invert	Changes the current species selection.
c	Edge length	Sets the displayed tile size.
d1 / d2	Rotation	Adjusts image rotation with a slider or numeric entry.
e	Flip LR	Flips the displayed images left-to-right.
f1 / f2	Mask below / above	Hides values outside the selected display bounds.
g1 / g2	Lower display limit	Sets the lower limit using a preset or custom value.
h1 / h2	Upper display limit	Sets the upper limit using a preset or custom value.
i	Link axes	Links zooming and panning across image tiles.
j	Auto rescale on zoom	Recomputes display limits from the currently zoomed image region.
k	Colormap	Chooses the image colormap.
l1-l7	Scale-bar controls	Toggles the scale bar and label, then sets style, location, length, font size, and color.
m1-m8	Despeckle controls	Enables artifact removal and selects method, kernel size, iterations, and spike threshold.
m9 / m10	Active-image indicator	A green check mark identifies the selected image tile.
n	Export	Exports the displayed maps as journal-style TIFF files using the current settings.

1. Select the dataset and species maps to display.
2. Adjust edge length to obtain a useful tile size.
3. Set quantile-based or custom lower and upper display limits.
4. Apply masking, rotation, or flipping when needed.
5. Link axes or enable automatic rescaling during detailed inspection.
6. Add and format a scale bar when physical coordinate information is available.
7. Optionally apply despeckling to reduce salt-and-pepper artifacts.
8. Export the displayed images when the presentation is ready.

4. Supporting GUIs

AutoSpect, RetroSpect, the Isotope Selection GUI, and ProSpect extend the core InSpect workflow. These tools are launched from the InSpect Programs menu or opened automatically during processing.

4.1 AutoSpect

AutoSpect provides semi-automatic processing and fitting of LA-ICP-TOF-MS imaging data acquired with supported TOFWERK instruments. It performs time-dependent peak-drift correction, Thomson recalibration, empirical peak-shape modeling, spectral deconvolution, and fitted-image reconstruction.

4.2 RetroSpect: Tailing Parameters

The Tailing Parameters tab is used to inspect and approve the empirical high- and low-m/z tail models used during AutoSpect fitting. The editable fields define the mass-dependent high-tail beta/sigma relationship, high-tail amplitude ratios, the low-tail beta/sigma constant, and the low-to-high tail amplitude relationship.

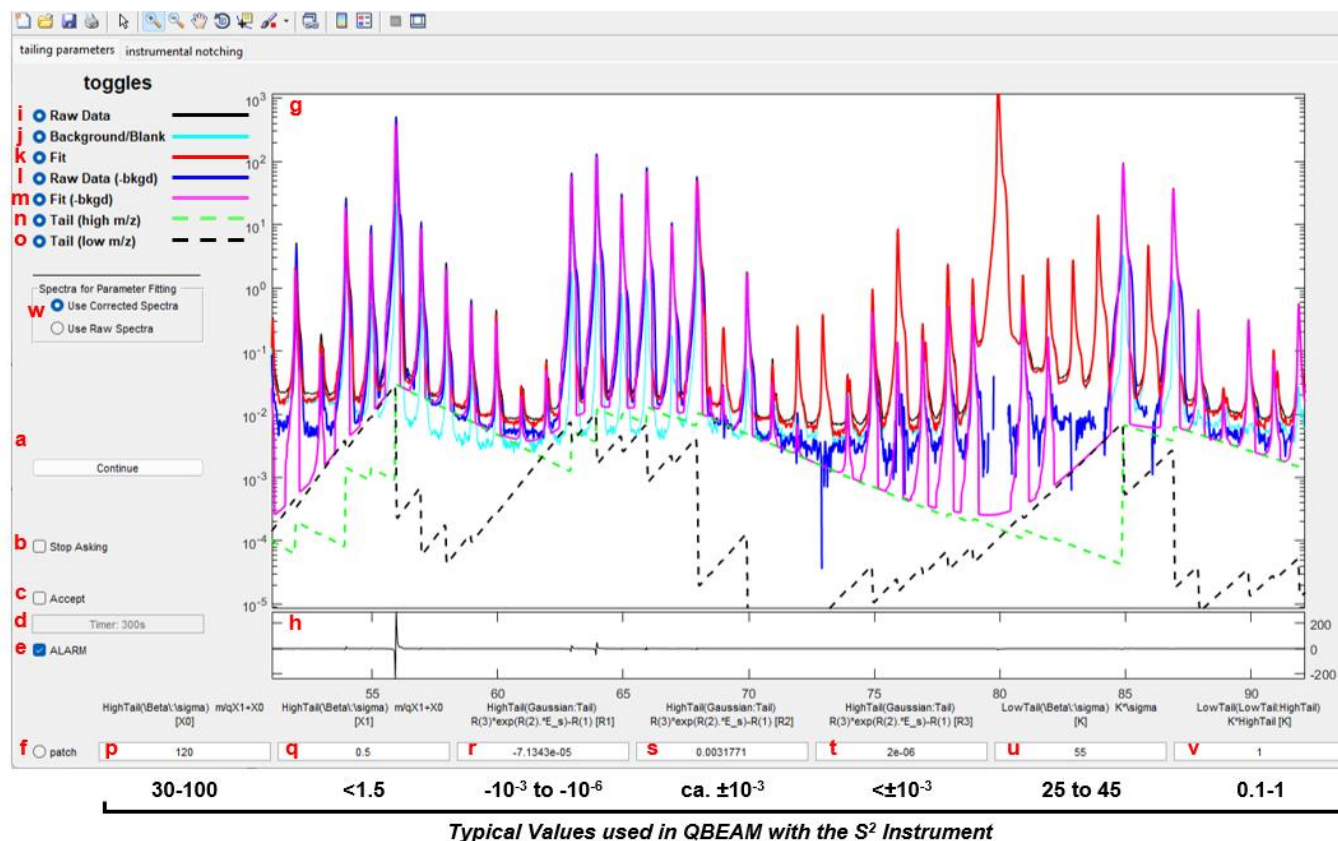


Figure 15. RetroSpect Tailing Parameters interface and parameter controls.

Callout	Control / Region	Function
a	Continue	Accepts the workflow state and proceeds to the notching interface.
b	Stop Asking	Enables batch operation without repeated approval prompts.
c	Accept	Uses the current parameter values when Continue is selected.
d	Timer	Counts down during fitting and proceeds automatically when it reaches zero.
e	Alarm	Produces periodic audible reminders while the interface is waiting for input.
f	Patch	Vestigial control; no routine user action is required.
g / h	Main plot / residual	Displays the fitted spectral components and the corresponding residual.
i-o	Display toggles	Toggles raw data, background/blank, fit, corrected data, corrected fit, and high- or low-m/z tails.
p / q	X0 / X1	Defines the first-order polynomial for the high-tail beta/sigma ratio as a function of m/z.

Callout	Control / Region	Function
r-t	R1 / R2 / R3	Defines the high-tail amplitude ratio as a function of m/z. V4.00+: For R1 (r) is controls the ratio of the areas and is constant. R2 and R3 (s and t) have been deactivated.
u	K: low-tail beta/sigma	Defines the constant low-tail beta/sigma ratio.
v	Low/high amplitude ratio	Defines the low-tail amplitude relative to the high-tail amplitude.
w	Spectra for Parameter Fitting	Chooses raw or background-corrected spectra for parameter determination.

4.3 RetroSpect: Instrumental Notching

The Instrumental Notching tab is used to review narrow m/z regions affected by instrument-related signal suppression. The position field defines the notch centroid, amplitude controls the multiplicative suppression correction, and FWHM defines the width of the Gaussian notch profile.

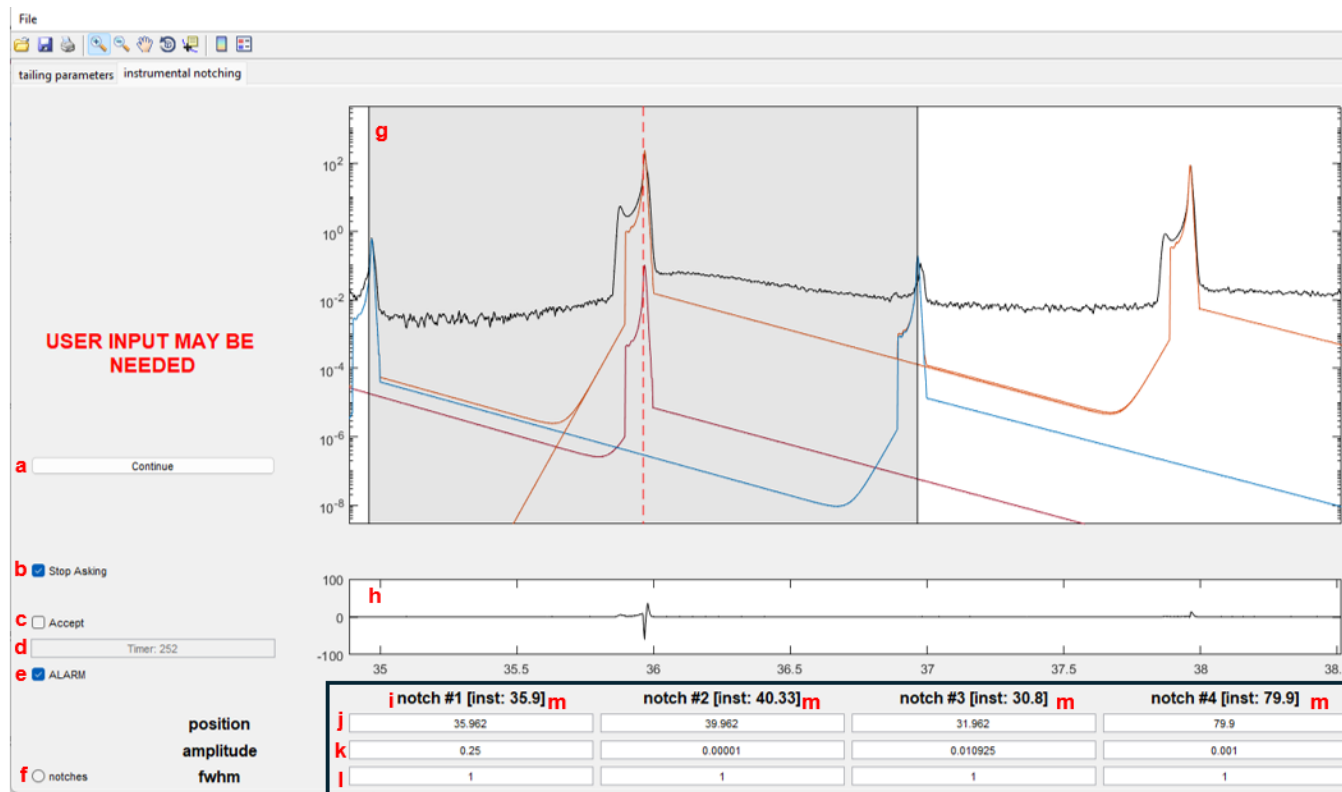


Figure 16. RetroSpect Instrumental Notching interface and notch controls.

Callout	Control / Region	Function
a-e	Continue / Stop Asking / Accept / Timer / Alarm	Operate as described for the Tailing Parameters tab.
f	Notches	Enables display of the notch model.
g / h	Main plot / residual	Displays the notched spectral model and the residual.
i	Notch-control groups	Contains the settings for each instrument notch.
j	Position	Sets the notch centroid in m/z.
k	Amplitude	Ranges from 0 to 1. A value of 1 applies no notching; lower values apply stronger suppression.
l	FWHM	Sets the notch width. Retain the workflow-specified value unless directed otherwise.
m	Approximate instrument values	Provides approximate m/z locations based on voltage settings. Right-clicking centers the plot around the entered notch position.

4.4 Isotope Selection GUI

The Isotope Selection GUI controls which isotopes, elements, and polyatomic ions are included in an AutoSpect spectral fit. The interface contains Isotope Table and Spectra tabs. Changes are propagated through the requested equation set and reflected in the model fit.

4.4.1 Isotope Table Tab

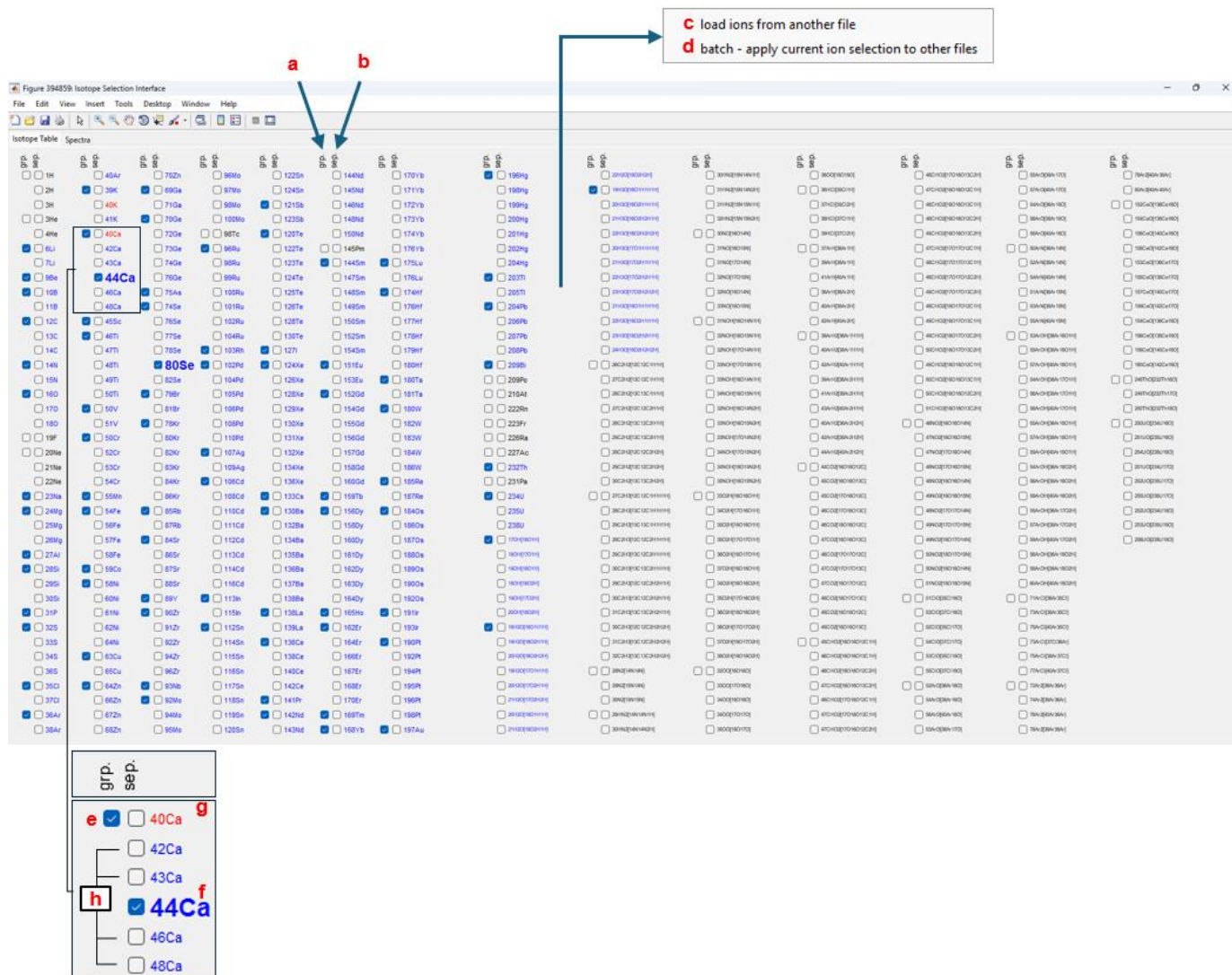


Figure 17. Isotope Table tab with group and separate-species controls and context-menu actions.

Callout	Control / Region	Function
a / e	grp.	Combines isotopes that are not treated separately or omitted, preserving branched fitting according to natural abundance.
b	sep. three-state control	Cycles an isotope through separate monoisotopic fitting, complete exclusion, and normal grouped fitting.
f	Separate state	Bold blue enlarged text indicates that the isotope amplitude is fitted independently of natural-abundance constraints.
g	Excluded state	Red text indicates that the isotope is omitted from fitting.
h	Grouped state	Small regular blue text indicates normal grouped/branched fitting when the group is enabled.
c	Load ions from another file	Loads an ion-selection pattern from another fitted dataset.
d	Batch apply ion selection	Applies the current ion selection to multiple processed files.

Note: The three-state separate-species control can require a short pause while the GUI updates. Click once, wait for the display to refresh, and then click again only when another state is required.

4.4.2 Spectra Tab

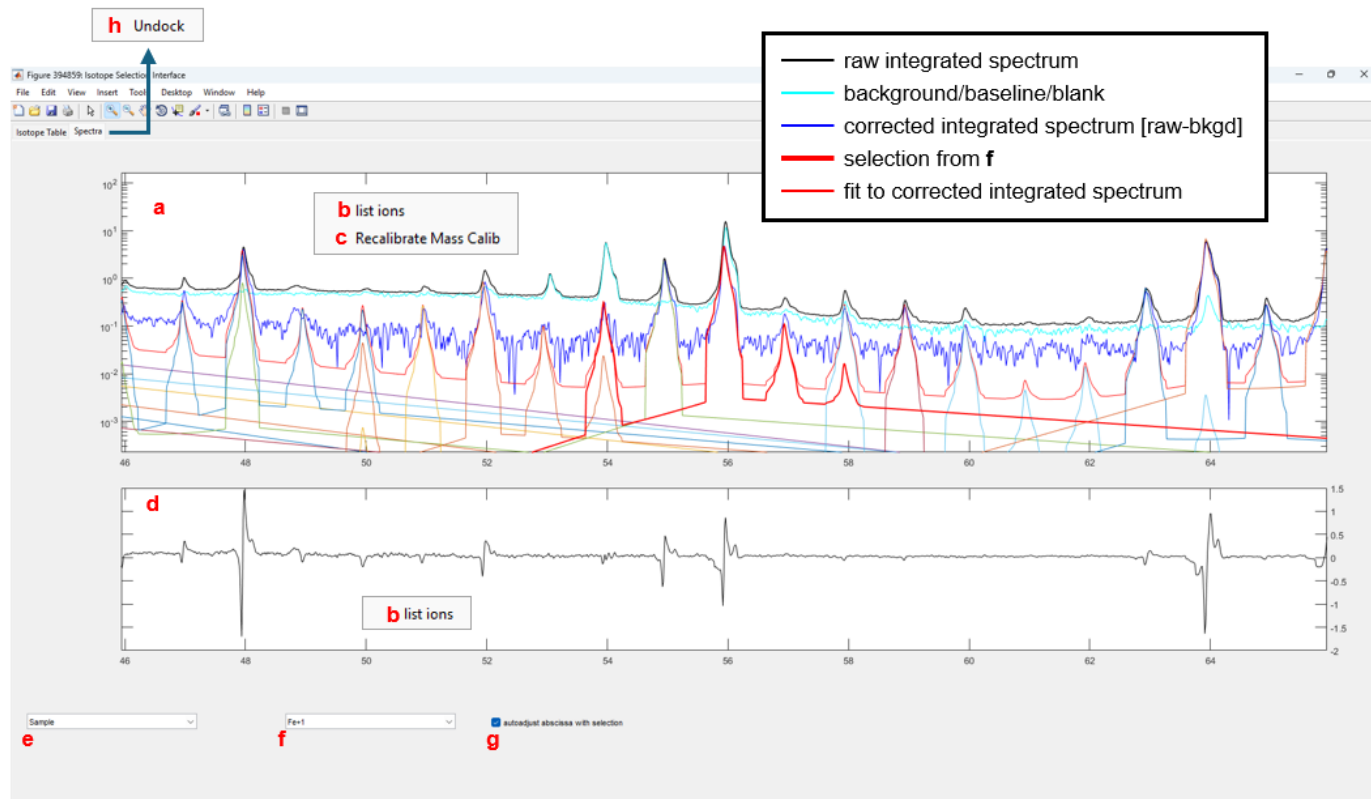


Figure 18. Isotope Selection GUI Spectra tab with fit components, residuals, spectrum selection, and isotope selection.

Callout	Control / Region	Function
a	Spectral component plot	Displays raw data, background, corrected data, modeled fit, and the selected species contribution.
b	List ions/isotopes	Select an m/z interval to list candidate species from the lookup table.
c	Recalibrate Mass Calibration	Development control; use only when the current workflow requires it.
d	Residual	Displays the residual of the current fit.
e	Spectra	Selects Sample, Gas Blank, or an individual reference spectrum.
f	Isotope / element / species	Selects the highlighted species shown in the spectral plot.
g	Auto adjust abscissa	Automatically centers the m/z display around the selected species.
h	Undock	Right-click the Spectra tab and select Undock for a separate window, useful on dual monitors.

Key	Action
Delete	Removes the current isotope or ion from fitting after confirmation and updates the Isotope Table.
Right arrow	Zooms the spectral x-axis in by approximately 10%.
Left arrow	Zooms the spectral x-axis out by approximately 10%.
Shift	Centers the spectral x-axis on the currently selected species.
Space	Resets the spectral x-axis limits to their defaults.

4.5 ProSpect

ProSpect monitors one or more directories for newly generated .h5 files. After a file is stable and no longer being written, ProSpect launches AutoSpect using the assigned RetroSpect parameter and isotope-selection files. A processing log records status, date, time, file name, file path, and the settings used.

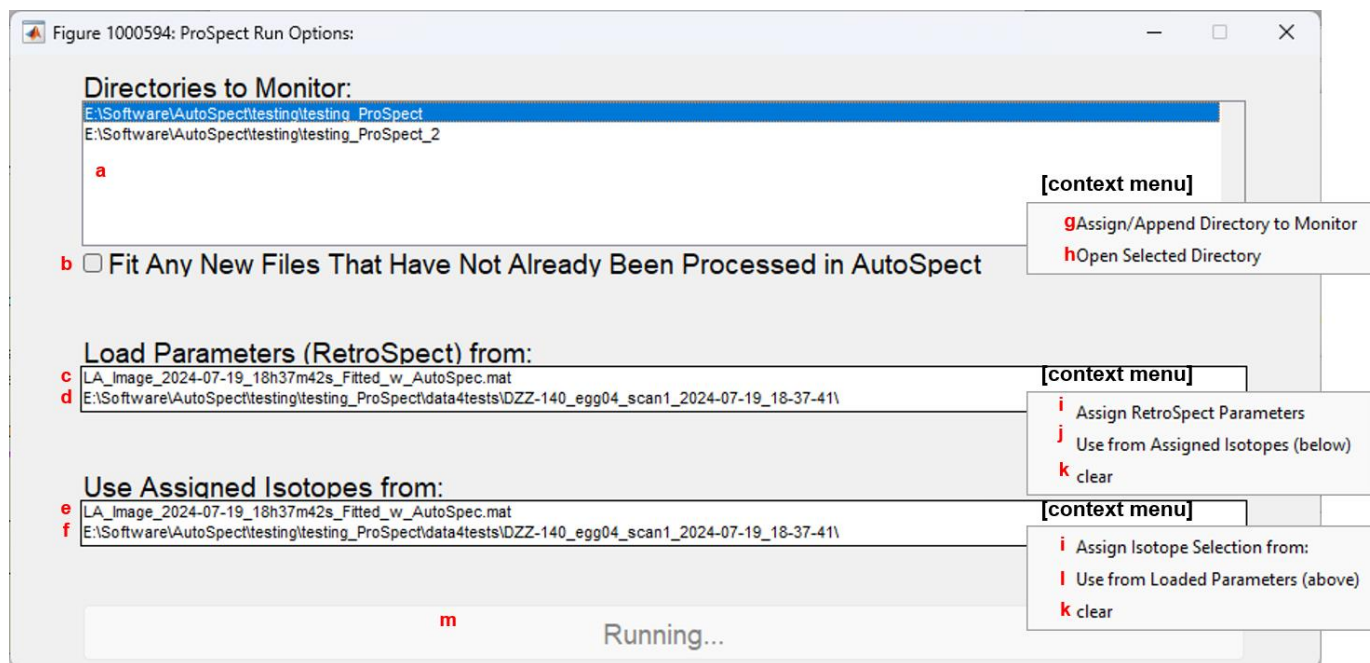


Figure 19. ProSpect run options and context menus for monitored directories, RetroSpect parameters, and isotope-selection files.

Callout	Control / Region	Function
a	Directories to Monitor	Lists the folders checked for new .h5 files.
b	Fit Any New Files Not Already Processed	Also processes existing files that do not already have a fitted AutoSpect output. When cleared, ProSpect waits only for files that appear after monitoring starts.
c / d	RetroSpect parameter file / path	Shows the assigned processing-parameter file and its directory.
e / f	Isotope-selection file / path	Shows the assigned isotope-selection file and its directory.
g	Assign/Append Directory to Monitor	Adds a directory to the monitored list.
h	Open Selected Directory	Opens the selected folder in the operating-system file browser.
i	Assign file	Assigns a RetroSpect parameter file or isotope-selection file in the corresponding context menu.
j	Use from Assigned Isotopes	Copies the assigned isotope-selection file/path into the RetroSpect parameter field.
k	Clear	Clears the corresponding assigned file field.
l	Use from Loaded Parameters	Copies the assigned RetroSpect parameter file/path into the isotope-selection field.
m	Run / Running	Starts directory monitoring. Closing the ProSpect window stops the monitoring timer.

5. Basic Workflows

5.1 Fit a New Dataset

1. Launch AutoSpect from InSpect.
2. Use the default ion selection or load a validated custom selection.
3. Select and confirm the raw files.
4. Review tailing and instrumental notching.
5. Allow AutoSpect to fit the data.
6. Review drift correction and fitted spectra in InSpect.
7. Save the fitted session.

5.2 Refine Isotope Selection

1. Open the Isotope Selection GUI from the Programs menu.
2. Use the Isotope Table to group, separately fit, or exclude species.
3. Use the Spectra tab to inspect the selected species and residuals.
4. Apply the current selection to additional fitted files only after the revised selection has been validated.

5.3 Build and Apply a Concentration Calibration

1. Load the fitted calibration dataset and reference-value table.
2. Review raster stability in Raster Filtering.
3. Assign each measurement to the correct standard.
4. Generate the forward and inverse models.
5. Exclude problematic rasters, series, or calibrants when justified.
6. Review smoothing, zero anchoring, residuals, monotonicity, and extrapolation.
7. Apply the accepted calibration to the loaded image dataset.

5.4 Export Publication-Style Images

1. Open the Images tab and select the desired dataset and species.
2. Set tile size, image orientation, colormap, and display limits.
3. Apply masking or despeckling only when appropriate for the intended output.
4. Add a scale bar and label when physical coordinates are available.
5. Select Export to save the current maps as TIFF images.

5.5 Run Unattended Processing with ProSpect

1. Add one or more acquisition directories.
2. Assign validated RetroSpect parameters and isotope-selection files.
3. Decide whether existing unprocessed files should also be fitted.
4. Select Run and leave the ProSpect window open while monitoring is required.
5. Review the ProSpect processing record for file status and settings.

6. Practical Notes and Checklist

Before Processing

- The raw files and acquisition directories are correct.
- The ion-selection file is appropriate for the current method.
- The selected tailing and notching parameters match the current instrument tuning.
- AutoSpect has write access to the output location.

Before Accepting a Spectral Fit

- The fitted spectrum overlaps the corrected measured spectrum across the relevant mass range.
- High- and low- m/z tails follow the baseline without crossing peak centroids.
- Instrumental notches align with the affected m/z regions and preserve appropriate isotope ratios.
- Residuals do not show strong structured errors around important peaks.

Before Applying a Concentration Calibration

- Reference assignments and concentrations are correct.
- Problem rasters, series, or calibrants are excluded only when justified.
- The forward surface is smooth and physically reasonable.
- The inverse model is monotonic over the usable range when an increasing response is expected.
- Residuals and diagnostics are acceptable for all required species.
- Extrapolation is limited and understood.

Before Exporting Images

- The selected dataset and species are correct.
- Display limits and masking are appropriate and reproducible.
- Rotation, flip state, aspect ratio, and scale bar are correct.
- Despeckling settings do not remove valid spatial features.
- The export format and destination are correct.

Tip: Once satisfactory tailing parameters have been established for a given instrument tuning, they can usually be reused for datasets acquired under similar tuning conditions.

Warning: Do not reuse tailing or notching parameters from substantially different instrument tuning conditions without reviewing the fit.

Advanced: The high- m/z tail is parameterized using a beta/sigma relationship that varies as a first-order polynomial with m/z ; mass-dependent amplitude ratios define the relative high-tail contribution.